

Mohammad Hasyim Taufik

Computational geoscientist
with expertise in scientific machine learning

hatsyim@gmail.com
+1 (346) 317-6642
<https://hatsyim.github.io/>

Education

- **King Abdullah University of Science and Technology** Thuwal, KSA
PhD in Machine Learning in Geosciences; GPA: 3.86 September 2025
- **King Abdullah University of Science and Technology** Thuwal, KSA
MSc in Machine Learning in Geosciences; GPA: 3.89 December 2021
- **Bandung Institute of Technology** Bandung, INA
BSc in Geophysics; GPA: 3.78 June 2019

Journal Articles

1. **M.H. Taufik** and T. Alkhalifah. High-Fidelity Velocity Model Building through Reconstruction-Guided Diffusion Model. *In preparation*.
2. **M.H. Taufik** and T. Alkhalifah. Full Waveform Inversion using Velocity-encoded Physics-Informed Neural Networks. *Under review*. Earth and Space Science, 2024.
3. **M.H. Taufik** and T. Alkhalifah. Wavenumber-aware Diffusion Sampling to Regularize Multi-parameter Elastic Full Waveform Inversion. *Published*. Geophysical Journal International, 2024.
4. **M.H. Taufik**, X. Huang, and T. Alkhalifah. Multiple Wavefield Solutions in Physics-Informed Neural Networks using Latent Representation. *Published*. IEEE Geoscience and Remote Sensing Letter, 2024.
5. **M.H. Taufik**, T. Alkhalifah, and U.B. Waheed. Stable Neural Network-based Traveltime Tomography using Hard-constrained Measurements. *Published*. Geophysics, 2024.
6. **M.H. Taufik**, and T. Alkhalifah. LatentPINNs: Generative Physics-Informed Neural Networks via Latent Representation Learning. *Accepted*. Artificial Intelligence in Geoscience, 2024.
7. **M.H. Taufik**, F. Wang, and T. Alkhalifah. Learned Regularizations for Multi-parameter Elastic Full Waveform Inversion using Diffusion Models. *Published*. JGR Machine Learning and Computation, 2024.
8. **M.H. Taufik**, U.B. Waheed, and T. Alkhalifah. A Neural Network Based Global Traveltime Function (GlobeNN). *Published*. Nature Scientific Report, 2023.
9. **M.H. Taufik**, U.B. Waheed, and T. Alkhalifah. Upwind, No More: Flexible Traveltime Solutions Using PINNs. *Published*. IEEE Transactions on Geoscience and Remote Sensing, 2022.

Conference Papers

1. **M.H. Taufik** and T. Alkhalifah. Efficient 3D velocity model building using conditional generative diffusion through reconstruction guidance. *Published*. EAGE Annual Meeting 2025.
2. **M.H. Taufik** and T. Alkhalifah. Accelerating Stein variational gradient descent in full waveform inversion with diffusion models. *Published*. EAGE Annual Meeting 2025.
3. **M.H. Taufik** and T. Alkhalifah. Conditionally guided diffusion sampling for enhanced North Viking Graben velocity model building. *Published*. EAGE Annual Meeting 2025.
4. S. Cheng, **M.H. Taufik**, and T. Alkhalifah. A generative neural operator for seismic wavefield representation. *Published*. EAGE Annual Meeting 2025.
5. S. Zhang, **M.H. Taufik**, and T. Alkhalifah. Robust FWI-Imaging by using diffusion model regularization. *Published*. EAGE Annual Meeting 2025.

6. **M.H. Taufik**, X. Huang, and T. Alkhalifah. Full-waveform inversion using velocity-encoded physics-informed neural networks. *Published*. SEG IMAGE Meeting 2024.
7. **M.H. Taufik**, F. Wang, and T. Alkhalifah. Diffusion regularization for multi-parameter near-surface elastic full-waveform inversion. *Published*. SEG IMAGE Meeting 2024.
8. **M.H. Taufik**, X. Huang, and T. Alkhalifah. Wavefield solutions using a physics-informed neural network as a function of Velocity. *Published*. EAGE Annual Meeting 2024.
9. **M.H. Taufik**, U. B. Waheed, and T. Alkhalifah. A stable neural network-based seismic tomography using hard-constrained measurements. *Published*. AGU Fall Meeting 2023.
10. **M.H. Taufik**, I. Yildirim, M. Ravasi, and T. Alkhalifah. Data and model hard constraints to physics-informed neural networks for near-surface tomography. *Published*. EAGE Annual Meeting 2023.
11. **M.H. Taufik**, T. Alkhalifah, and U. B. Waheed. A robust seismic tomography framework via physics-informed machine learning with hard-constrained data. *Published*. EAGE Annual Meeting 2023.
12. **M.H. Taufik**, U. B. Waheed, and T. Alkhalifah. Upwind, no more: flexible traveltime solutions using physics-informed neural networks. *Published*. EAGE Annual Meeting 2022.
13. **M.H. Taufik**, U. B. Waheed, and T. Alkhalifah. The Eikonal solution for attenuating VTI media using physics-informed neural networks. *Published*. EAGE Annual Meeting 2021.
14. **M.H. Taufik**, U. B. Waheed, and T. Alkhalifah. Eikonal solution using physics-informed neural networks for global seismic travel time prediction. *Published*. AGU Fall Meeting 2021.

Experience

- **Occidental Petroleum Corporation** Houston, USA
Subsurface Innovation Lab Intern *December 2024 - February 2025*
 - **Uncertainty Quantification in Inverse Problems:** Develop an AWS-based solution to quantify the risk of misplacing borehole locations using seismic data.
 - **Diffusion Models:** Accelerating the solutions of non-linear inverse problems in AWS using conditional diffusion models.
- **KAUST** Thuwal, KSA
MS/PhD Fellow *August 2020 - Present*
 - **Physics-Informed Neural PDE Solvers:** Develop a deep learning-based neural partial differential equation (PDE) solvers.
 - **Diffusion Regularization for Inverse Problems:** Pioneer the use of a diffusion model to regularize multi-parameter non-linear inverse problems.
- **Cloud4C** Jakarta, INA
Machine Learning Engineer Intern *June 2022 - August 2022*
 - **Computer Vision:** Built and deployed a machine learning model to detect safety equipments given a set of images on Amazon AWS.
 - **Natural Language Processing:** Built web application based on Flask for a web summarizer given a text, website link, or PDF file as input.
 - **Cloud Architecture:** Deployed machine learning model on Amazon AWS and orchestrated the data IO between S3, the user's mobile phone, and the local machine for a recommendation system application.
- **Pertamina Hulu Kalimantan Timur** Balikpapan, INA
Geophysicist *June 2019 - May 2020*
 - **Interpretation:** Involved in the regional joint study along the Makassar Strait.
 - **Geophysics:** Generating well-seismic tie process, seismic data quality assessment, depth-time conversion process.
 - **Geology:** Generating three-dimensional (3D) velocity model to build 3D pore pressure cube.
 - **Reservoir:** Initiated rock physical data visualization for sand-shale distribution analysis.

- **Bandung Institute of Technology**

Bandung, INA

Research Assistant

Auguts 2017 - June 2019

- **Seismic Processing:** Constructing a guideline in seismic data processing and basic signal processing. Mentoring and grading the course learning process.
- **Seismology:** Constructing a guideline in basic seismology course. Supervising basic seismological routine processing like earthquake relocation and arrival time picking.
- **Seismic Refraction:** Constructing a guideline in exploration seismic refraction course. Supervising routine seismic refraction processing routine.
- **Geophysical Instrumentation:** Supervising basic electrical engineering behind potential methods in geophysics, electrostatic, and electrodynamic in geophysics. Mentoring and grading the course learning process.
- **Computational Geophysics:** Constructing a guideline in basic numerical methods in geophysical prospecting methods. Supervising basic numerical methods for simple gravimetry methods.
- **Electrical Analysis:** Constructing a guideline in basic electrical engineering and signal processing in geophysics. Supervising basic electrical engineering behind potential methods in geophysics, electrostatic, and electrodynamic in geophysics.

- **PetroChina**

Jakarta, INA

Geophysicist Intern

June 2018 - August 2018

- **Quantitative Seismic Interpretation:** Analyzing elastic properties of reservoir rocks through rock physics analysis.
- **Rock Physics:** Generating well-seismic tie process and enhancing rock properties distribution map through multi-attribute analysis.

Projects

- **Physics-informed Neural Networks (PINNs):** Solving a parametric PDE for 2D to 3D Earth's velocity model for regional to global Earth scale.
- **Mass Spectrometry:** Data visualization web application based on plotly with data processing in R.
- **Wireline Transformers:** Wireline logs prediction using a Transformers-based machine learning model.
- **Multi-parameter Land Data Elastic Full-waveform Inversion using Deep Learning Regularization:** Introduce novel regularization scheme using deep generative (diffusion) models to invert for V_p , V_s , ρ simultaneously from single-component geophones.

Talks

1. **Student Presenter.** EAGE Annual Meeting. Oslo, NOR. June 2024.
2. **Guest Lecture.** Bandung Institute of Technology. Bandung, INA. December 2023.
3. **Student Presenter.** AGU Annual Meeting. New Orleans, USA. December 2023.
4. **Student Presenter.** EAGE Annual Meeting. Vienna, AUT. June 2023.
5. **Student Presenter.** SEG IMAGE Meeting. Houston, USA. September 2022.
6. **Student Presenter.** EAGE Annual Meeting. Madrid, ESP. June 2022.
7. **Student Presenter.** AGU Annual Meeting. New Orleans, USA. December 2021.
8. **Student Presenter.** EAGE Annual Meeting. Amsterdam, NED. October 2021.
9. **Student Presenter.** IPA Annual Convention and Exhibition. Jakarta, INA. September 2019.

Honors, Awards, and Fellowships

- **Dean's Award** 2025
King Abdullah University of Science and Technology
- **NVIDIA-EAGE HPC Hackathon Winner** 2024
European Association of Geoscientists and Engineers
- **Best Teamwork in the EVOLVE Program** 2022
Society of Exploration Geophysicists
- **Middle East Super Bowl Champion** 2021
Society of Exploration Geophysicists
- **KAUST Fellow** 2020 - Present
King Abdullah University of Science and Technology
- **Dean's Honor List** 2015 - 2019
Bandung Institute of Technology
- **Best Geosciences Student Presenter** 2019
Indonesian Petroleum Associations
- **Best Thesis Presentation** 2019
Bandung Institute of Technology
- **Best Student Paper Presenter** 2018
Trisakti University

Programming Skills

- **Languages:** Matlab, Python, R, Bash, LaTeX, C++
- **DevOps:** AWS, Docker, Kubernetes, GCP
- **Machine Learning:** TensorFlow, Keras, PyTorch, Scikit-learn, SciANN
- **Data Visualization:** D3.js, Plotly, Carto
- **Front-end:** Hugo, CSS, HTML5
- **Databases:** SAS, PostgreSQL, SQLite, MySQL, MongoDB, BigQuery, Apache Spark